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Inverse design on customised absorption of acoustic metamaterials with high degrees of freedom by deep learning

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ABSTRACT

On-demand design of acoustic metamaterials remains challenging due to iterative, resource-intensive trial-and-error processes. This study presents a novel conditional generative adversarial network (CGAN)-based inverse design method for metaporous materials with perforated plates (MMPP), enabling rapid and efficient generation of embedded inclusions within metaporous materials in a high-freedom design space. Leveraging prior acoustic knowledge, a custom dataset incorporating closed rigid and thin-walled inclusions is constructed to support broadband, low-frequency absorption. The CGAN learns the nonlinear mapping between target broadband absorption spectra and MMPP configurations, achieving high fidelity (mean squared error (MSE) $= 5.03 \times 10^{-3}$, mean absolute percentage error (MAPE) $= 6.39\%$) between the predicted absorption curves of inverse designs and the target absorption curves from the test set. Unlike conventional data-driven methods, this CGAN-based inverse design method integrates structural constraints and physics-guided post-processing to generate manufacturable, interpretable, and novel designs beyond the training data distribution. The generated patterns corresponding to different acoustic performances reveal underlying physical correlations: structures optimised for low-frequency absorption generally exhibit lower filling ratios and shallow-embedded inclusions, whereas those designed for high-frequency performance tend to feature higher filling ratios and deep-embedded inclusions. Furthermore, the method produces diverse structure candidates tailored to user-defined absorption targets. With a thickness of 51 mm, the inverse designs achieve effective sound absorption (α>0.5) across 360–3000 Hz. Multiple absorption peaks are induced through a combination of micro-perforation resonance, local resonance of embedded inclusions, and higher-order modal resonances coupled with acoustic wave interference. This work enables rapid, performance-driven inverse design and provides a new approach to exploring the structure-performance coupling mechanisms in acoustic metamaterials.

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1. Introduction

Acoustic metamaterials [1–4] have emerged as an effective solution for resolving noise among researchers due to their unique acoustic properties. Although traditional sound-absorbing porous materials [5–7] can effectively dissipate mid-to-high frequency sound energy, it is challenging to achieve low-to-mid frequency sound absorption within a compact structure. One approach to enhancing the sound absorption performance of porous materials is the development of metaporous materials, where rigid inclusions are embedded within the porous matrix. This design couples the viscous and thermal losses of porous materials with additional absorption mechanisms. The shape and dimensional parameters of these rigid inclusions have a significant impact on the sound absorption properties [8–12], necessitating careful design to achieve the desired acoustic performance.

The traditional design method for acoustic metamaterials is to adjust design parameters based on the researcher's prior experience and physics-inspired methods. Forward modelling methods, such as the finite element method (FEM) and the transfer matrix method (TMM) [13], are used to iteratively adjust design parameters until the design goal is achieved. Despite numerous successful implementations of traditional design methods [14–17], it is inherently cumbersome, time-consuming, and heavily dependent on the intuition of the researcher. Additionally, the design parameters and design space are limited. Furthermore, since the effects of various sub-structures on sound absorption performance are coupled, optimising each parameter individually often does not yield the optimal design.

Through the introduction of optimisation techniques, important progress has been made in the design of acoustic metamaterials. Many scholars have employed optimisation algorithms, such as the genetic algorithm (GA) [18–20] and the particle swarm optimisation (PSO) [21–23], to identify optimal structural parameters for obtaining target performance. Guo et al. [21] combined the PSO with theoretical prediction models to design the optimal multi-layer resonators with extended necks, optimising their performance over the prescribed frequency range. Similarly, Gao et al. [18] designed an acoustic metamaterial composed of nickel foam, a lateral board, and a back cavity, and a GA was applied to optimise geometric parameters. Their approach maximised the average absorption coefficient over a broad frequency range (200 Hz–20 kHz), considering both normal incidence and oblique sound wave incidence.

Nevertheless, given the broad possibilities for fabricating acoustic metamaterials with complex geometries and intricate spatial distributions of material properties, the space of possible design parameters is vast. Although many optimisation algorithms, such as GA and PSO, have achieved success in various applications, their performance can be influenced by the specific characteristics of the search space and the algorithmic parameters. Techniques that rely on predefined rules and specific search steps may encounter limitations in efficiently exploring high-dimensional design spaces. The further development of computational models and design strategies for acoustic materials is required.

Advancements in computer hardware technology have made calculations fast and economical, and accelerated the acquisition process of data sets, enabling broader applications of existing deep learning methods. More and more researchers have begun to focus on how to combine deep learning with metamaterial design, leading to the emergence of acoustic metamaterial inverse design [24–28]. The concept of inverse design is to directly determine an optimal structure that meets specified dynamic performance criteria, bypassing traditional physics-based design approaches. Compared to traditional approaches, inverse design often explores a broader design space, which can increase the complexity and challenges of the design process. A key aspect of inverse design is the effective representation of structural configurations and the collection of a large dataset of structural samples to support the learning process.

There are two primary strategies for structure characterisation. The first involves designing structural configurations defined by specific parameters and generating a dataset by systematically varying these parameters [29–32]. For instance, Xiao et al. [31] adopted deep learning to realise a coherently coupled metasurface absorber containing nine parallel Helmholtz resonators, which accelerates the parameter optimisation process. Song et al. [33] integrated analytical models and numerical models to obtain a large training dataset for laminated plate-type acoustic metamaterials, and realised diverse designs for target frequency bands using deep learning. However, as the number of design parameters increases, the training accuracy of the model tends to decrease, thereby constraining the design flexibility of acoustic metamaterials. To address this limitation in high-dimensional design spaces, the autoencoder model (AE) [34] and the variational autoencoder-based model (VAE) [35,36] can be employed to perform nonlinear dimensionality reduction, effectively mapping high-dimensional design parameters onto a lower-dimensional latent space.

The second strategy involves pixelating the structural pattern, representing it in matrix form, and using the spatial distribution of matrix values to encode the material distribution of the structure [37–42]. This pixel-based representation enables greater structural diversity and design flexibility. Pan et al. [41] developed an optimisation framework integrating a convolutional neural network (CNN) with a GA to guide the design of metaporous materials by inserting rigid partitions. In their approach, metaporous material patterns were described as binary images with a resolution of 30×50 pixels.

For the design scene with a high-degree-of-freedom design space, it is necessary to use networks that are good at processing high-dimensional designs, such as Transformers and generative adversarial network (GAN) [43,44], to ensure that the design accuracy is within an acceptable range. GAN, first introduced by Goodfellow et al. [45], achieved a significant breakthrough in generative modelling by learning the underlying data distribution. Subsequently, various extensions and modifications were proposed to address issues such as training instability and generation quality, including the conditional generative adversarial network (CGAN) [46]. Many studies have leveraged the advantages of GAN and its variants in the field of image generation to explore the vast design space of material. Li et al. [47] utilised the GAN to effectively generate failure samples in the high-dimensional design space in topology optimisation, thereby accelerating the optimisation process and enhancing the efficiency of finding the optimal topology. Donda et al. [48] designed an acoustic metasurface with embedded apertures and labyrinth channels, mesh-encoded the solid and air regions into a pattern matrix, and then, using GAN-generated candidate designs that match a target frequency of 82 Hz.

Although recent advancements in deep learning have proven to be highly beneficial in discovering new patterns in various fields,

the application of inverse design for the on-demand design of high-degree-of-freedom acoustic metamaterials aimed at sound absorption performance needs to be further explored. To expand the sound absorption bandwidth within the dataset, this study introduces metaporous materials covered with a perforated plate (MMPP), guided by prior knowledge. An inverse design method based on a CGAN is proposed to generate the distribution topology of the MMPP. Moreover, image processing plays a critical role in inverse design. Following dataset collection, images undergo connected domain analysis; after the inverse design is completed, they are further refined through filtering and smoothing. This work is organised as follows: Section 2 details the dataset construction process, including the design details of MMPP patterns, corresponding sound absorption curve calculations, as well as the theoretical background and simulation models for sound absorption. Section 3 focuses on the CGAN architecture and training processes employed for the inverse design of MMPP. Section 4 discusses the post-processing methods for MMPP design images, evaluates the prediction performance of the trained CGAN, presents the relationship between the size and distribution of inclusions and the sound absorption curve, and shows the inverse design results based on custom broadband sound absorption curves. Section 5 provides experimental validation of the sound absorption performance of the configurations obtained through inverse design. Finally, Section 6 concludes the study and provides future perspectives.

2. Dataset collection

Traditional design methods for on-demand acoustic metamaterials are not only time-consuming but also struggle to achieve optimal designs. To address this limitation, data-driven deep learning algorithms are adopted to accelerate the design process. In such approaches, the target frequency range for inverse design relies on the dataset composition. Therefore, constructing a dataset that includes acoustic metamaterials with both low-frequency and broadband sound absorption capabilities is essential to enable greater design flexibility. Improving the dataset by introducing prior knowledge from established acoustic design strategies [6,49,50], which includes: (1) using porous materials with low or medium flow resistivity as base units for broadband sound absorption, (2) embedding rigid inclusions into porous materials to construct metaporous materials and enhance global sound absorption, and (3) introducing perforated plates to achieve low-frequency sound absorption. For simplicity and capability in material design, the continuous material

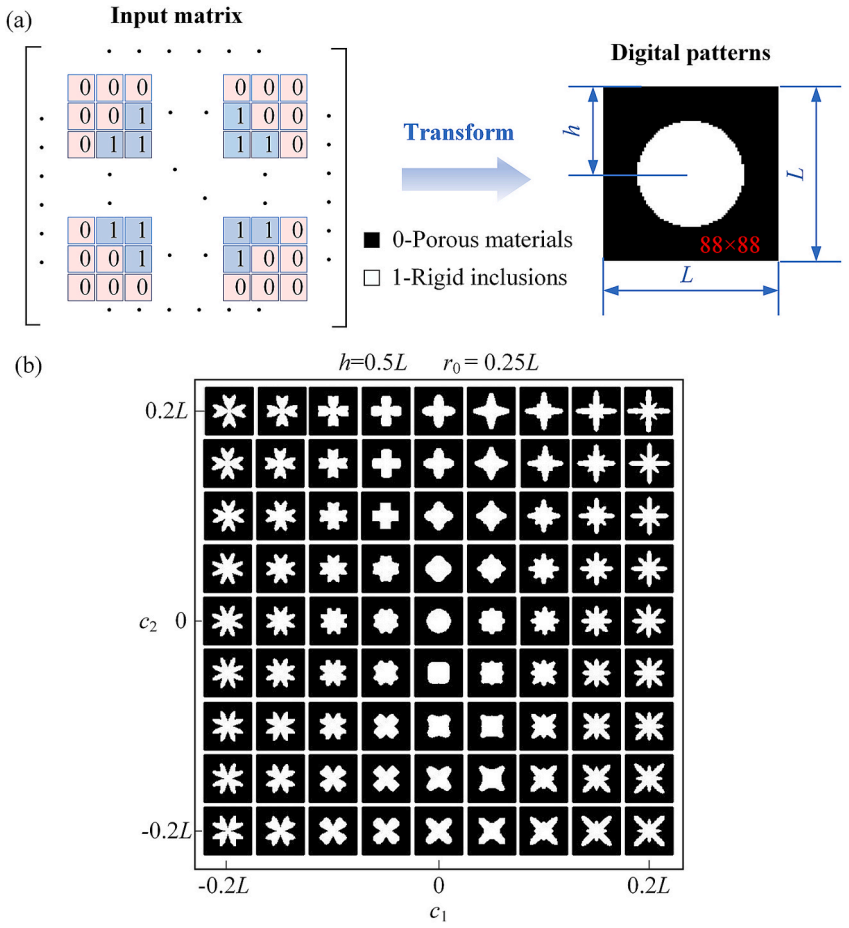


Fig. 1. (a) 0/1 distribution matrix and corresponding digital patterns (0/1 denotes porous materials/rigid inclusions); (b) Schematics of porous materials embedded with rigid inclusion with different c_1 and c_2 (black parts and the white parts represent porous materials and rigid inclusions).

domain is discretised into finite binary elements using digital material methods.

2.1. Digital pattern generation strategy

The original acoustic metamaterial design concept in this work mainly consists of metaporous materials and a perforated plate, in which metaporous materials are porous materials embedded with rigid inclusions.

The dataset consists of digital patterns representing MMPP with two distinct material components. As shown in Fig. 1(a), the 88 × 88 pixels binary images represent the cross-section of the MMPP, where '0' (black pixels) denotes porous material and '1' (white pixels) denotes rigid inclusions. The images are designed with a relatively high resolution, facilitating the representation of complex structures and offering an expanded design space. However, excessively high resolutions can substantially increase computational costs. To balance computational efficiency and structural accuracy, a resolution of 88 × 88 is chosen. This resolution provides sufficient detail to enable the global design of acoustic metamaterials with a high degree of freedom. The image generation process is implemented using Python.

For MMPP with a porous material matrix of fixed dimension, the shape and the size of rigid inclusions are critical to the sound absorption performance. It is necessary to ensure the diversity and representativeness of patterns in the dataset as much as possible. Therefore, two types of inclusion patterns are included in the dataset: closed-rigid inclusion and thin-walled inclusions, both embedded in porous materials. All structural patterns are generated under manufacturability and functionality constraints to avoid unreasonable designs.

2.1.1. Closed-rigid inclusion embedded in porous materials

Regarding the closed-rigid inclusion, it is a closed shape without openings. According to previous works [9,10], the filling fraction and barycenter position of closed-rigid inclusion in metaporous materials significantly influence acoustic absorption frequency. To control the boundaries between rigid inclusions and porous materials, closed B-spline curves are adopted. These curves allow for variable shape, filling fraction, and barycenter position of rigid inclusions, and their mathematical representation in polar coordinates is given as:

$$\begin{cases} x = r\cos(\theta) \\ y = r\sin(\theta) + 0.5L - h \\ r = r_0(1 + c_1\cos(4\theta) + c_2\cos(8\theta)) \end{cases} \tag{1}$$

where x and y denote the Cartesian coordinates. The parameter θ represents the angular parameter spanning $[0, 2\pi]$, which ensures the inclusion maintains an integral closed B-spline boundary. The key parameters governing the inclusion geometry are:

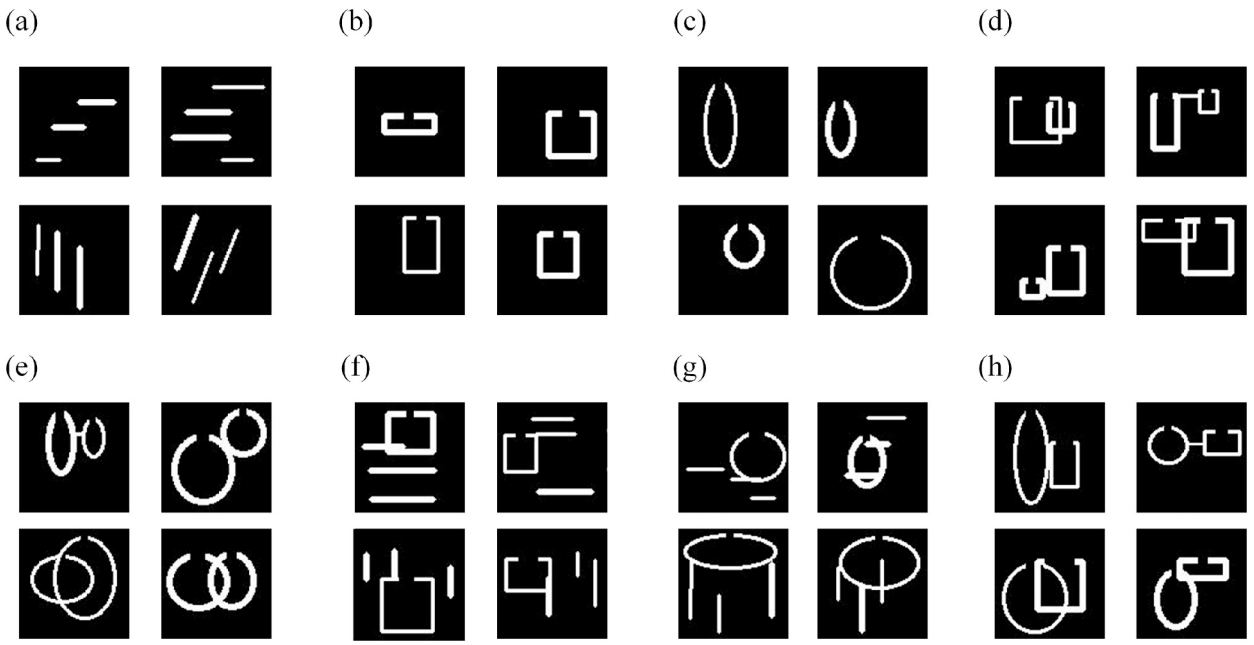


Fig. 2. Typical patterns of thin-walled rigid inclusions embedded in porous materials with partitions, rectangular-shaped cavities, and C-shaped cavities: (a) metaporous materials with partitions; (b) metaporous materials with a rectangular-shaped cavity; (c) metaporous materials with a C-shaped cavity; (d) metaporous materials with double rectangular-shaped cavities; (e) metaporous materials with double C-shaped cavities; (f) metaporous materials with a rectangular-shaped cavity and partitions; (g) metaporous materials with a C-shaped cavity and partitions; (h) metaporous materials with a rectangular-shaped cavity and a C-shaped cavity.

h: Adjusts the barycenter position of the inclusion within the porous matrix.
r_0: Determines the overall size of the region enclosed by the B-spline curve.
c_1 and c_2: Jointly control the shape of the rigid inclusion.

The size of the porous matrix is set as $L \times L$, as displayed in Fig. 1(a). To ensure practical feasibility, the key parameters are constrained within reasonable ranges:

$$\begin{cases} h \in [0.3L, 0.1L, 0.7L] \\ r_0 \in [0.1L, 0.05L, 0.25L] \\ c_1 \in [-0.2L, 0.05L, 0.2L] \\ c_2 \in [-0.2L, 0.05L, 0.2L] \end{cases} \tag{2}$$

To generate diverse image samples, four key parameters (h, r_0, c_1, c_2) are discretised into equal steps as presented in Eq. (2) and systematically combined. Fig. 1(b) illustrates examples of patterns where barycenter position $h = 0.5L$ and the size$r_0 = 0.25L$ remain fixed, while c_1 and c_2 vary, demonstrating different inclusion shapes. This parametric approach enables the efficient and large-scale generation of diverse patterns, optimising the dataset for deep learning applications.

2.1.2. *Thin-walled rigid inclusion embedded in porous materials*

For patterns denoting thin-walled rigid inclusions embedded in porous materials, the control parameters of the pattern are obtained by a random function within the physical constraints, and the randomly obtained parameters are combined to generate the pattern. Thin-walled inclusions include three types of basic elements: partitions, rectangular-shaped cavities, and C-shaped cavities. For pattern diversity, these elements are combined either as single elements or in pairs. Examples of these configurations are shown in Fig. 2, where Fig. 2(a)–(c) plot those patterns with a single element, Fig. 2(d)–(e) present patterns with a combination of the same element, Fig. 2(f)–(h) display patterns with a combination of different elements.

These types of inclusions exhibit local resonance characteristics. By controlling the spacing between partitions, the size of the cavity openings, and the overall cavity dimensions, the sound absorption frequency can be tuned. Thus, to ensure the functionality and manufacturability of the generated patterns, specific constraints are applied:

(1) Boundary clearance: To prevent inclusions from covering the entire horizontal and vertical span of the pattern, a minimum clearance of $0.045L$ is reserved at the top, bottom, left, and right boundaries of the porous matrix, which provides the necessary cutting allowance for porous materials processing;
(2) Wall thickness: Based on structural stability and lightness requirements, the minimum and maximum wall thicknesses for basic elements are $0.03L$ and $0.06L$, respectively;
(3) Partition spacing: To maintain effective acoustic wave propagation channels and ensure printing quality, the distance between the nearest points of the individual partitions is set to a minimum of $0.1L$;
(4) Cavity size: Following Helmholtz resonator design principles, the size of the rectangular-shaped cavity ranges from $0.15L$ to $0.5L$, and the size of C-shaped cavity ranges from $0.1L$ to $0.4L$. The minimum opening width is set to $0.034L$ based on printing precision limitations, and the maximum opening width is limited to 1/3 of the minimum length (for rectangular cavities) or short axis (for C-shaped cavities) to ensure acoustic performance.

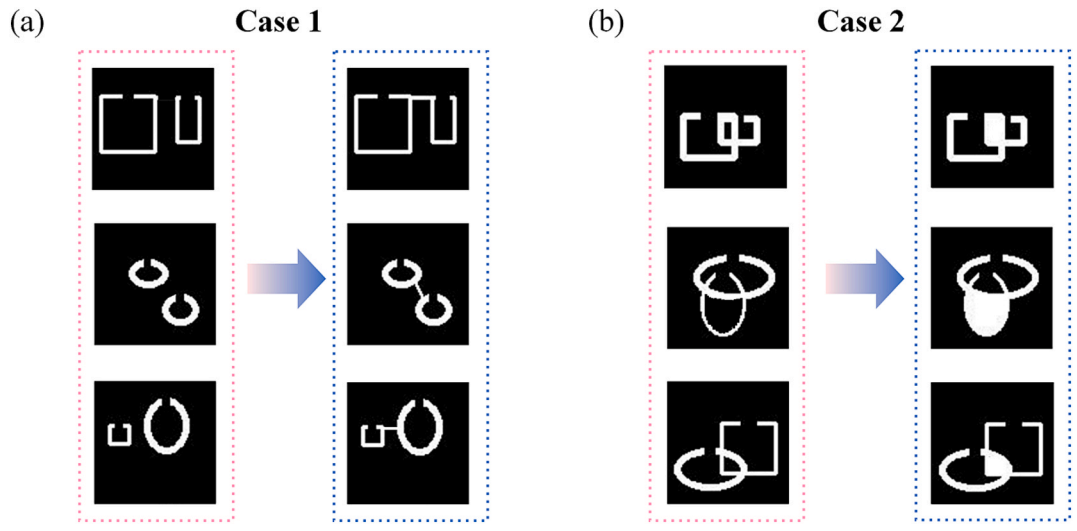


Fig. 3. Schematic of pattern preprocessing: (a) Case 1: patterns with two isolated elements (excluding those with partitions element); (b) Case 2: patterns with two connected elements.

Before simulation, a pattern preprocessing step is applied to simplify the computational domain. This preprocessing aims to reduce the number of discrete domains in the patterns, which is essential for efficient numerical calculations. Specifically, patterns with multiple elements are reprocessed in two cases:

Case 1 (Isolated elements): When the two elements are isolated (excluding those with partition elements), a partition with a width of $0.04L$ and a length equal to the minimum distance between the isolated elements is added to connect them, as shown in Fig. 3(a).

Case 2 (Connected elements): When the elements are connected and form a closed domain, pixel inversion is performed on the closed areas formed by the connections, as shown in Fig. 3(b).

The pattern processing in Case 1 is applied to both the dataset and numerical simulations, ensuring consistency between the training data and the computational model.

For Case 2, closed regions formed by the connection of multiple elements are filled to create closed-rigid inclusions, emphasising the necessity of including patterns with closed-rigid inclusions in the dataset. While this filling reduces the geometric complexity of the simulation model, it also increases pattern complexity and diminishes characteristic resonant cavity features, potentially affecting the effectiveness of pattern feature extraction in the inverse design network.

To balance these considerations, two versions of Case 2 patterns are maintained: the original patterns with connected elements are preserved in the dataset to retain structural geometry properties, while the processed versions with filled closed areas are used in simulations for computing sound absorption spectra. This approach ensures computational efficiency without altering the acoustic response of the original patterns, thereby maintaining the validity of the simulation results.

2.2. Acquisition of acoustic absorption spectra for digital patterns

This study primarily focuses on the design of sound absorption performance for acoustic metamaterials. To achieve this, deep learning is employed to model the relationship between digital patterns and sound absorption performance. Consequently, the sound absorption spectra corresponding to the structural configurations represented by the digital patterns are of great concern.

2.2.1. Acoustic absorption theory background

In this section, a theoretical foundation for understanding the mechanisms of sound absorption in multilayer systems is provided. The proposed acoustic metamaterials MMPP mainly consists of two components: a perforated plate and metaporous materials, and it is theoretically analysed from two aspects. First, the specific acoustic impedance Z_{pp} of the perforated plate [51] is determined by

$$Z_{pp} = \rho_0 c_0 (r + \mathrm{j}\omega m), \tag{3}$$

where ρ_0 is the air density, c_0 is the sound velocity in air, r is the relative acoustic resistance, and m is the relative acoustic mass.

Second, the specific acoustic impedance Z_p of porous material is given by

$$Z_p = \sqrt{\rho_e K_e}, \tag{4}$$

where ρ_e is effective dynamic mass density and K_e is effective bulk modulus. The porous material in metaporous materials with rigid backing is modelled by the Johnson-Champoux-Allard (JCA) model [52,53] with five crucial acoustical parameters: porosity ϕ, tortuosity α_∞, flow resistivity σ, viscous length Λ and thermal characteristic length Λ'. ρ_e and K_e are given by

$$\rho_e = \frac{\rho_0 \alpha_\infty}{\phi}\left[1 + \mathrm{j}\frac{\omega_\nu}{\omega}F(\omega)\right], \tag{5}$$

$$K_e = \frac{\phi}{\gamma P_0}\left(\gamma - (\gamma - 1)\left[1 + \mathrm{j}\left(\frac{\omega_c'}{Pr\omega}\right)G(Pr\omega)\right]^{-1}\right)^{-1}, \tag{6}$$

where $\omega_\nu = \sigma\phi/\rho_0\alpha_\infty$ is angular Biot frequency, $\omega_c' = \sigma'\phi/\rho_0\alpha_\infty$ is the adiabatic/isothermal crossover angular frequency and thermal resistivity $\sigma' = 8\alpha_\infty\eta/\phi\Lambda'^2$, γ is the specific heat ratio, P_0 is the atmospheric pressure, Pr is the Prandtl number, and the $F(\omega)$ and $G(Pr\omega)$ are the correction function, which is determined by

$$F(\omega) = \sqrt{1 - \mathrm{j}\eta\rho_0\omega\left(\frac{2\alpha_\infty}{\sigma\phi\Lambda}\right)^2}, \tag{7}$$

$$G(Pr\omega) = \sqrt{1 - \mathrm{j}\eta\rho_0 Pr\omega\left(\frac{2\alpha_\infty}{\sigma'\phi\Lambda'}\right)^2}, \tag{8}$$

where η is the viscosity of the fluid with $\mathrm{j}^2 = -1$.

By using the transfer matrix method [54], the acoustic impedance of the system Z_s is obtained as

$$\boldsymbol{T}_{total} = \boldsymbol{T}_{PP} \cdot \boldsymbol{T}_P = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}, \tag{9}$$

$$Z_s = T_{11}/T_{21}, \tag{10}$$

where $\boldsymbol{T}_{pp}$ and $\boldsymbol{T}_p$ are the transfer matrix of a perforated plate and porous materials and are given by

$$\boldsymbol{T}_{pp} = \begin{bmatrix} 1 & Z_{pp} \\ 0 & 1 \end{bmatrix}, \tag{11}$$

$$\boldsymbol{T}_p = \begin{bmatrix} \cos(k_e h) & -\mathrm{j}Z_p \sin(k_e h) \\ \dfrac{\mathrm{j}}{Z_p}\sin(k_e h) & \cos(k_e h) \end{bmatrix}, \tag{12}$$

Then, the reflection coefficient r in the sound field and absorption coefficient α [55] of the system can be calculated as

$$r = \frac{Z_s - \rho_0 c_0}{Z_s + \rho_0 c_0}, \tag{13}$$

$$\alpha = 1 - |r|^2 \tag{14}$$

This theoretical framework encompasses the effect of micro-perforated plates and porous materials on the absorption coefficient, which facilitates the subsequent understanding of the absorption coefficient curve simulated by the numerical model.

2.2.2. *Acoustic absorption spectra obtained by numerical model*

After obtaining sample patterns, the corresponding sound absorption spectra of the sample patterns also need to be collected to construct the dataset. This dataset allows the deep learning model to learn the relationship between patterns and their acoustic absorption spectra. For this purpose, the finite element method (FEM) is adopted in COMSOL Multiphysics 6.0 to evaluate the sound absorption spectra of different patterns. During the FEM simulation process, the sound pressure distribution is expressed by the - Helmholtz equation [56], which can be written as follows:

$$\nabla \cdot \left(-\frac{1}{\rho}(\nabla p - q_d) \right) - \frac{\omega^2 p}{\rho c^2} = Q_m, \tag{15}$$

where ρ and c are density and sound velocity, p denotes the sound pressure, q_d and Q_m represent the dipole domain source and monopole domain source, respectively.

Since the generated sample patterns are pixel-based images, they need to be converted into geometries that can be recognised and processed by COMSOL Multiphysics for finite element (FE) simulation. It is worth noting that MATLAB is used for image preprocessing and command automation, while all calculations of sound absorption coefficients are performed entirely in COMSOL. The integration and interconnection between MATLAB and COMSOL are realised through COMSOL Multiphysics LiveLink™ with MATLAB module, which ensures an automated and efficient simulation process.

Since the cross-sectional geometry of the MMPP model remains constant along the vertical direction, a two-dimensional (2D) simulation model can be employed instead of a three-dimensional one to simplify the computational process, as illustrated in Fig. 4(a).

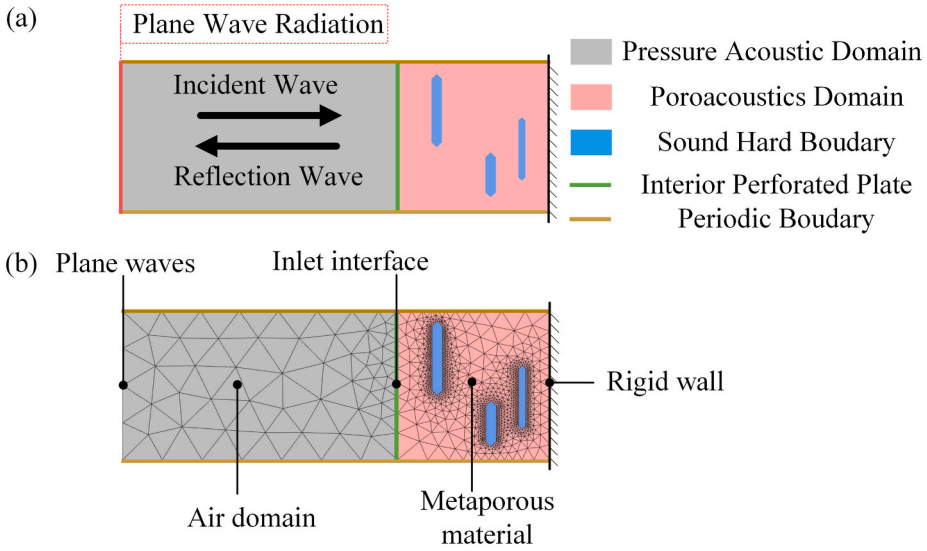


Fig. 4. Illustration of (a) a two-dimensional FE model, (b) and FE mesh.

An internal perforated plate interface (green solid line) is applied, with the plate thickness, hole diameter, and perforation rate set to 1 mm, 1.5 mm, and 0.06, respectively. The metaporous materials are designed with a thickness and width of 50 mm to meet the requirements of compact engineering applications. Periodic boundary conditions illustrated by brown lines are applied to both sides of the computational domain. To reduce the mesh count while maintaining computational accuracy, a free triangular mesh is used with a maximum element size that is smaller than one-sixth of the minimum wavelength, as shown in Fig. 4(b).

Melamine foam [57] is chosen as a porous matrix of metaporous materials, due to its excellent sound absorption performance, ease of processing into various shapes, and wide application in noise control and architectural acoustics. The corresponding JCA parameters of melamine foam are listed in Table 1.

Subsequently, the frequency domain study is conducted with a range of (40, 40, 3000) Hz to evaluate the absorption coefficient of the inspected MMPP. The simulated acoustic response is a collection of sound absorption coefficients measured at 75 discrete frequency points. This simulation requires approximately 16 s of computation time on a standard AMD Ryzen 7 5800H processor.

In particular, the sound absorption spectra of four representative generated patterns are analysed using a numerical model: porous materials embedded with a closed circular inclusion, porous materials embedded with partition plates, porous materials embedded with rectangular-shaped cavities, and porous materials embedded with C-shaped cavities. Additionally, the sound absorption performance of basic configurations, including homogeneous porous materials, individual perforated plates, and porous materials with perforated plates, is compared with that of representative generated patterns to illustrate the advantages of the generative design strategy, as displayed in Fig. 5.

Fig. 5(a) suggests that the results of the numerical analysis align well with the theoretical results, confirming the accuracy of the numerical model. Homogeneous porous materials show great sound absorption performance above 1000 Hz, while a perforated plate with an air cavity has an absorption peak with a relatively low absorption coefficient above 1000 Hz. Moreover, Fig. 5(b) demonstrates that introducing a perforated plate into a homogeneous porous material weakens the mid-to-high frequency sound absorption while shifting the absorption peak to lower frequencies, thereby broadening the effective sound absorption range. Especially, the different rigid inclusions, including closed inclusion and thin-walled inclusion within porous materials covered by a perforated plate, can adjust the second sound absorption peak, as observed in Fig. 5(c)–(f). This adjustment enhances mid-to-high frequency absorption, effectively compensating for the absorption loss introduced by the perforated plate in this frequency range. These findings confirm that the proposed design strategy of embedding various inclusions within porous materials covered by a perforated plate is both feasible and effective in enhancing sound absorption performance.

2.3. Dataset composition

After collecting 14,420 digital patterns with 88 × 88 pixels, the corresponding acoustic absorption coefficients at 75 frequency points (40–3000 Hz) for each pattern are calculated by performing 14,420 individual FE simulations. Notably, these simulations are executed continuously, without manual intervention for each pattern. The dataset, shown in Fig. 6(a), consists of 14,420 samples, each including a binary image with 88 × 88 pixels and its corresponding absorption spectra with 1 × 75 frequency points. To ensure effective deep learning training, both the dataset size and the distribution of different inclusion features are carefully controlled. Therefore, the nine types of digital patterns, representing different inclusion features, are balanced as evenly as possible.

The absorption coefficient distribution within 40–3000 Hz for 14,420 patterns is shown in Fig. 6(b). It reveals that the effective sound absorption ($\alpha > 0.5$) range is from 256 Hz to 3000 Hz. The considered sound absorption spans from low to medium frequencies (40–2000 Hz), with consideration also given to the high-frequency band of 2000–3000 Hz. This range encompasses the dominant frequency components of traffic noise, construction noise, and general engineering noise [58–60]. In contrast, noise above 3000 Hz can be effectively attenuated using conventional methods, such as homogeneous porous materials [6], which are already well-established and widely applied. Therefore, to balance model complexity and computational cost, the frequency range considered in the dataset construction is limited to 40–3000 Hz. The created dataset is comprehensive and can meet the needs of most application scenarios.

Although the construction of the dataset requires a considerable number of finite element simulations, it represents a one-time computational investment. Once the subsequent deep learning model training is completed, the customised metamaterial structure can be predicted within seconds without further simulation. Therefore, the initial simulation effort greatly enhances the efficiency, scalability, and flexibility of the subsequent design process, making it highly valuable for practical applications involving iterative design tasks, large design spaces, or multiple target requirements.

3. CGANs for inverse design of MMPP

The core of the inverse design method lies in the construction of CGAN, which is trained using the collected dataset to learn both the distribution of patterns and their nonlinear relationship with the sound absorption coefficient.

Table 1
Acoustical parameters of melamine foam.

ϕ	α_∞	σ(Pa · s · m^{-2})	Λ(μm)	Λ'(μm)
0.98	1.01	11,401	39.8	335.8

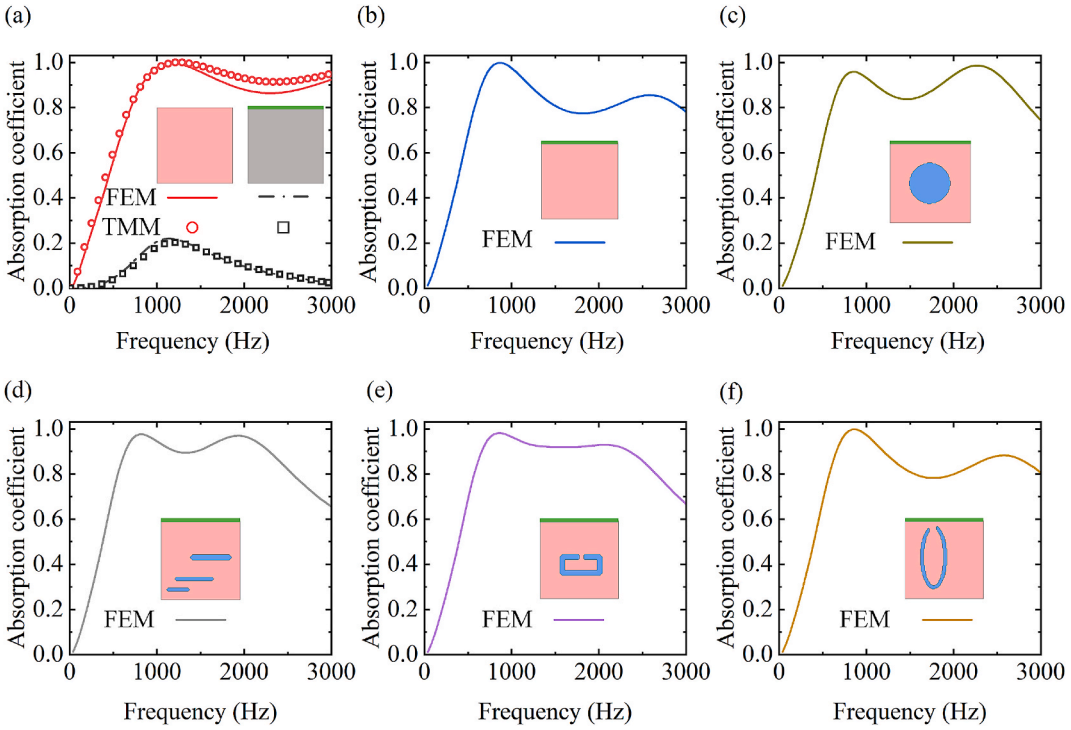


Fig. 5. Sound absorption spectra of (a) homogeneous porous, perforated plate with air cavity (lines represent FEM results, dots represent TMM results), (b) perforated plate with porous materials (the green boundary denotes the interior perforated plate boundary), (c) and typical patterns with perforated plate in dataset, such as porous materials embedded with closed circular inclusion, (d) porous materials embedded with partition plates, (e) porous materials embedded with rectangular-shaped cavities, (f) porous materials embedded with and C-shaped cavities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.1. Workflow and mechanism of CGANs

In this section, the deep learning algorithm, which constitutes the core part of the inverse design method, is examined. GAN [45], an unsupervised learning algorithm, comprises two modules: a generator (G) and a discriminator (D), enabling automatic image generation. The main training process of GAN is two networks competing iteratively to continuously update the weights, until reaching a Nash equilibrium.

As for the classic GAN, it mainly estimates the distribution of image samples, so as to generate images similar to those in the training data. However, CGAN [46], a GAN variant, can incorporate an additional condition to control the generation of the corresponding image. Hence, CGAN is suitable for acoustic metamaterials design tasks, involving sound absorption requirements. In this work, the additional condition is sound absorption spectrum.

The schematic workflows of GAN and CGAN are presented in Fig. 7. The CGAN training process follows a systematic procedure similar to GAN. First, the parameters of G and D are initialised. Subsequently, a batch is created by extracting a small portion of the training dataset. The G generates images based on conditional information, with initially generated images generally being noisy. Then, two-dimensional images and their corresponding acoustic absorption spectra are extracted from the training dataset, and mixed with images carrying additional conditions generated by the G. These mixed images are fed into D to determine whether each image is from the generator or the real dataset, during which the parameters of G remain fixed. The parameters of D are continuously updated until the image is correctly identified as real or fake, where “fake” images are generated from G and “real” images are drawn from the dataset.

If the D can correctly identify the source of the image, it stops updating and fixes the parameters. After this process, the parameters of G are updated via the backpropagation algorithm. After multiple loop iterations, the stronger the classification ability of the D is, the stronger the deception ability of the G is until a balance between the two is reached. Both the G and the D are trained using a binary cross-entropy loss function [61] as given as follows:

$$E_{BCE} = -\frac{1}{N}\sum_{k=1}^{N} y_k \log(p_k) + (1 - y_k)\log(1 - p_k), \tag{16}$$

where N denotes the size of the output layer, y_k is the label value of the sample, and p_k is the probability that the D outputs a binary label value y_k. The goal of the D is to minimise the loss function, while the goal of the G is to maximise the loss function.

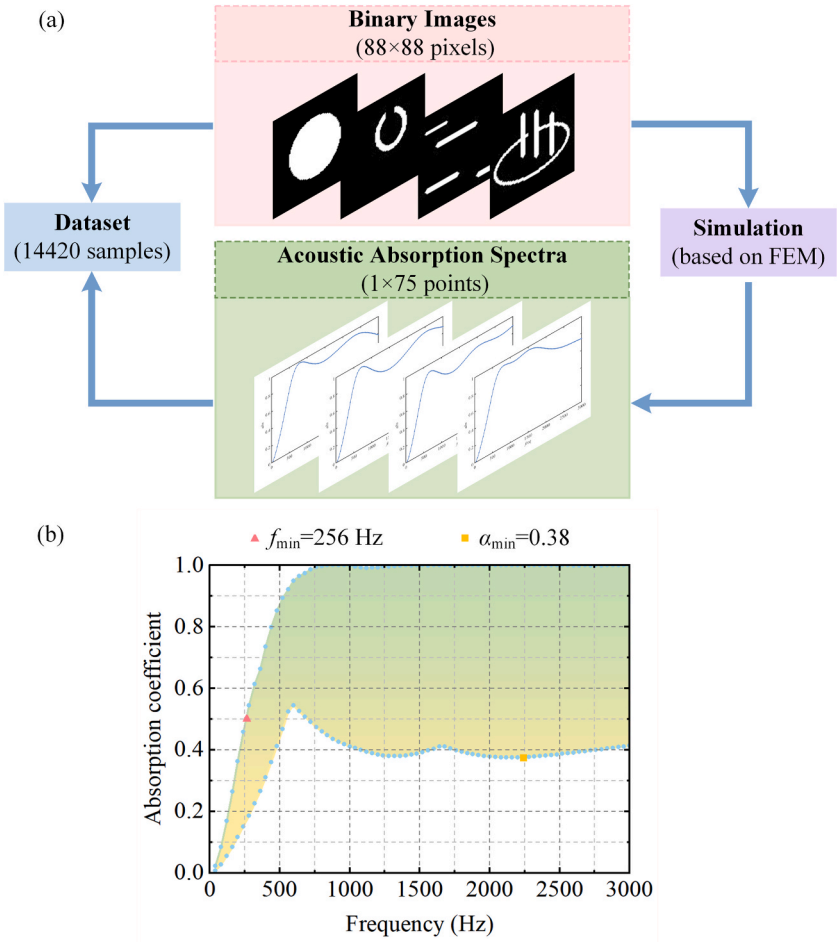


Fig. 6. (a) Flowchart of data set acquisition with 14,420 digital patterns and corresponding acoustic absorption spectra; (b) Distribution of sound absorption coefficient for two-dimensional MMPP in the dataset.

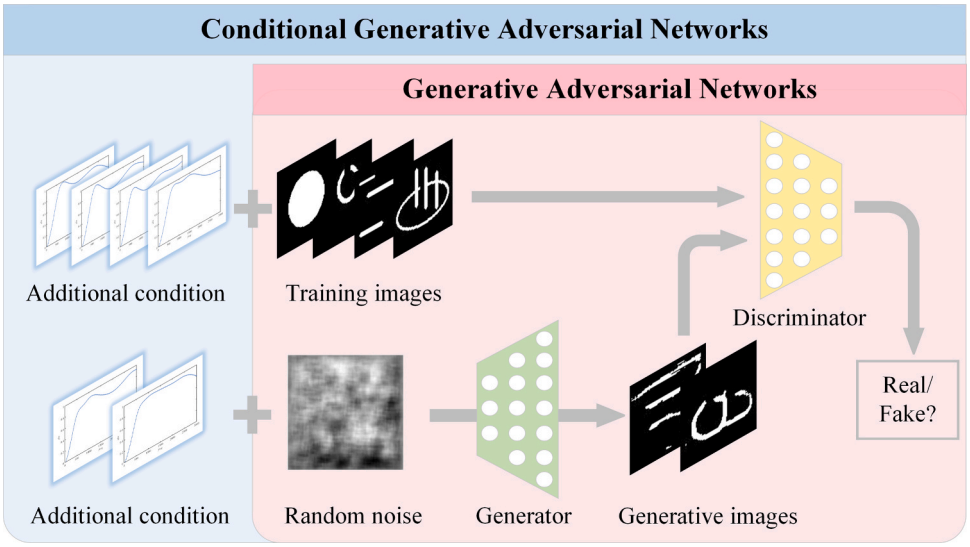


Fig. 7. Schematic of the training of GAN and CGAN.

Next, the network architecture of CGAN and the training details are discussed.

3.2. Network architecture of CGAN

G and D are composed of convolutional neural networks, which are particularly effective in processing two-dimensional data such as images. The detailed network architecture of G is introduced with five convolutional layers as exhibited in Fig. 8(a). There are two input channels in G, one processes the 1×75 acoustic absorption spectra from the dataset, while the other introduces 1×128 Gaussian noise. This random noise serves as a source of variability and a tool for capturing the underlying data distribution, enabling the generator to create diverse and unique outputs even from similar input spectra. Moreover, it lays the foundation for the generalisation capability of the G and the effectiveness of the training process. Before the two inputs are merged into a single layer, both are processed separately through fully connected dense layers and then reshaped into the desired image format. Then, one merged layer is processed by five convolutional layers utilising 5×5 kernels. Upsampling layers are used after every convolutional layer to increase the dimension of the input to learn the appropriate weight. The output of G is a two-dimensional image, which is an 88×88 pixel matrix. The Leaky Rectified Linear Unit (LeakyReLU) activation function is applied in the first four layers, where LeakyReLU is expressed by

$$\mathrm{LeakyReLU}(z) = \begin{cases} z & z \geqslant 0 \\ 0.2z & z < 0 \end{cases} \tag{17}$$

The activation function of the last layer is the Hyperbolic Tangent Function (Tanh), as given by

$$\mathrm{Tanh}(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}} \tag{18}$$

The LeakyReLU helps mitigate the vanishing gradient problem, enhancing feature extraction capability and training efficiency. Meanwhile, the G aligns its output with the real data processed by the D through the Tanh activation function. Thus, this combination of activation functions improves the overall training performance [47,62]. The batch normalisation is implemented to stabilise the

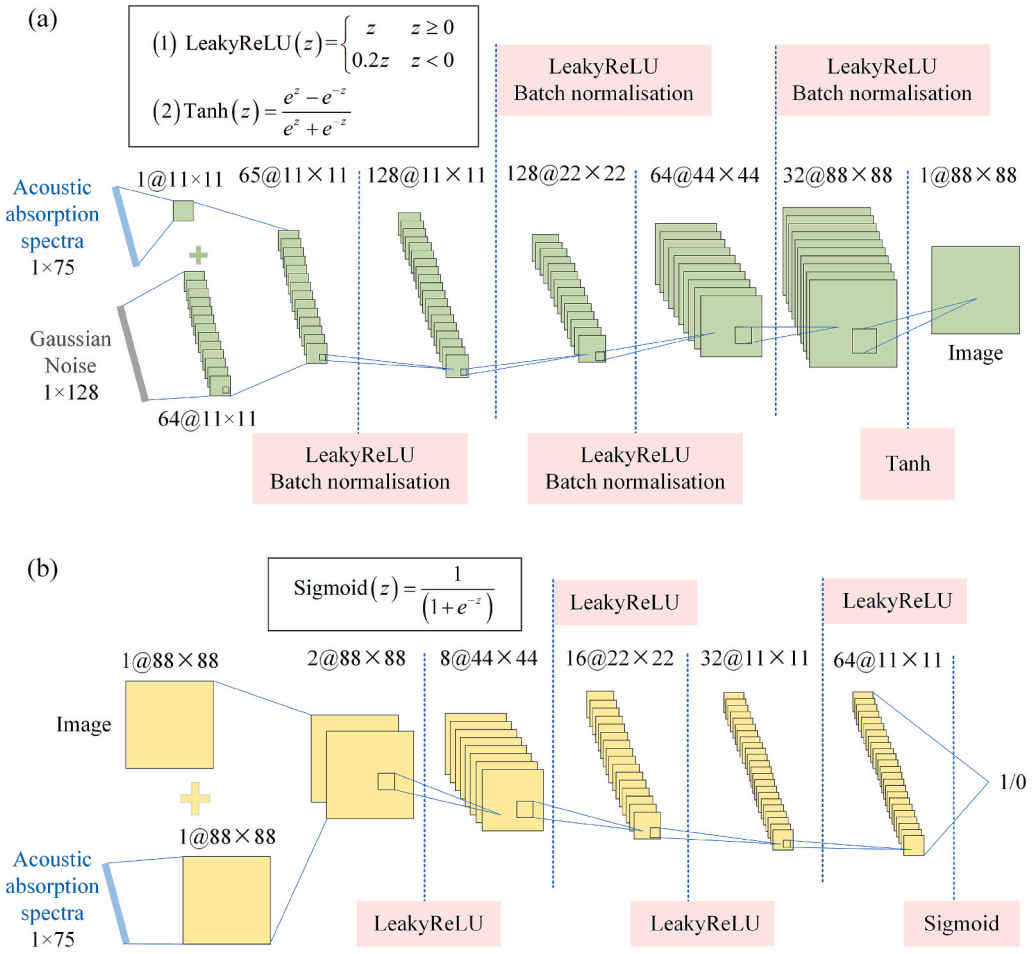


Fig. 8. Detailed network architecture of CGAN: (a) schematics of a generator (G); (b) schematics of a discriminator (D).

training process and avoid overfitting.

The D contains two input channels as well, the first one is for 88 × 88 images to be classified as input, and the other is to input the 1 × 75 acoustic absorption spectra accompanying images, as shown in Fig. 8(b). Since the acoustic absorption spectra 1 × 75 is in a one-dimensional format, they need to be reshaped into a two-dimensional image format and connected to an image to be classified. Hereafter, the connected input is processed by four convolution layers utilising 5 × 5 kernels. The output of D is a category label value, it classifies the image as "real" or "fake" and performs 1 or 0, respectively. The first three layers of the convolutional network apply the activation function LeakyReLU, and the last layer uses the Sigmoid activation function as given by

$$\mathrm{Sigmoid}(z) = \frac{1}{1 - e^{-z}} \tag{19}$$

The output value of the Sigmoid function is limited to 0 and 1, which meets the output value requirements of the D.

When the D and G have a similar number of layers, adversarial training may become insufficient, potentially leading to mode collapse. Conversely, if the G is excessively deep, the D may overfit, diminishing its ability to distinguish real from fake samples, which can also contribute to mode collapse. Therefore, a well-balanced learning dynamic between the G and the D is crucial for ensuring stable training. This study is conducted on a system with an AMD Ryzen 7 5800H CPU, NVIDIA GeForce RTX 3050 Laptop GPU (4 GB/6GB VRAM), and 16 GB RAM. Considering computational constraints and drawing insights from existing literature on GAN training [47,48,63,64], the final network architecture is established through an iterative trial-and-error process.

3.3. Training details of CGAN

3.3.1. Data preprocessing

Before training the neural network, data preprocessing is crucial for optimizing model performance. Therefore, a multi-step preprocessing strategy is implemented to enhance training efficiency.

The original image has a value range [0,1], where the prevalence of zero values is detrimental to the full activation of the first layer of network and consequently extends training duration. To address this limitation, the value range is transformed to [−1,1] by replacing 0 with −1, and this process is displayed in Fig. 9(a). This substitution accelerates network convergence. Additionally, the [−1,1] distribution aligns with the symmetric output of the final layer of G using Tanh activation function, ensuring consistency in data representation.

To further enhance model robustness, the dataset is randomly shuffled to rearrange the sequence of various pattern types. The dataset, consisting of 14,420 samples, is then split into a training set and a test set with a ratio of 0.85:0.15. The training set is used for model training, while the test set is reserved for the final evaluation of model performance. Standardisation is applied to the sound absorption spectral data to improve training efficiency and prevent overfitting caused by any inherent order in the dataset.

3.3.2. Hyperparameter settings

The training process employs the Adam optimiser, with the learning rate of the G set to 0.0001 and that of the D set to 0.0004. This differentiated learning rate effectively balances the optimisation speeds of the two networks, thereby improving training stability [65]. The training consists of 10,000 iterations with a batch size of 32. With limited computational resources, a larger batch size generally facilitates more accurate gradient estimation. In the G, convolutional layers utilise filter counts of 128, 128, 64, 32, and 1, respectively, with a stride of 1. In the D, convolutional layers also employ filter counts of 8, 16, 32, and 64, with strides of 2, 2, 2, and 1, respectively. To prevent overfitting, dropout regularisation is applied in each layer of D with an update rate of 60%. It effectively prevents overfitting without losing crucial information needed for the accuracy of D. The above hyperparameter settings are obtained by gradient traversal search guided by prior research insights [47,48,63,64].

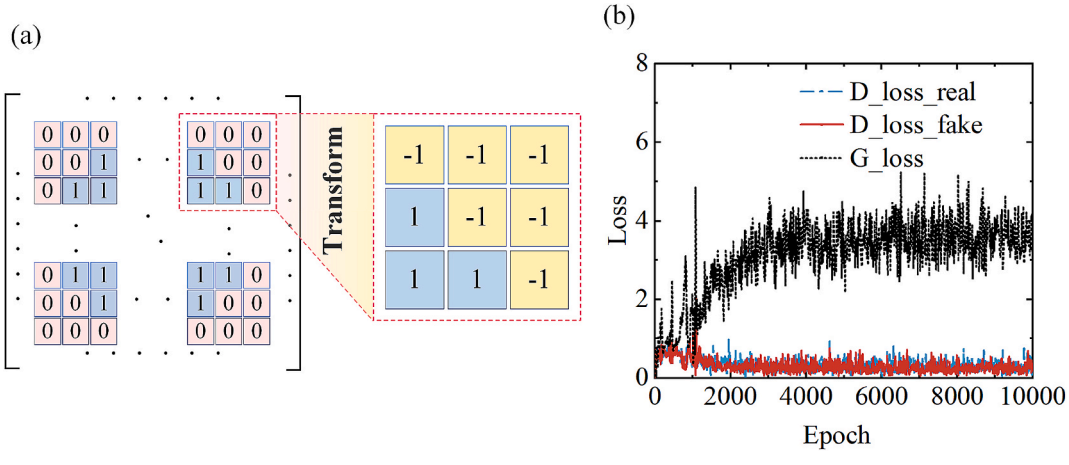


Fig. 9. (a) Schematic diagram of image data preprocessing; (b) Loss plot for CGAN training.

The training process is shown in Fig. 9(b). During the first 1,000 iterations, the loss curve exhibits significant fluctuations. The loss is defined according to Eq. (16). Afterwards, the loss curve gradually stabilises. The loss of the G settles between 3 and 5, while the loss of the D approaches 0 for both real and fake data. The stability of the loss of the G and the D indicates that the two have reached a balance, which means that the training has achieved the expected success.

4. Results and discussion

Once the CGAN model is successfully trained, it can not only capture the data characteristics of the input images but also learn the nonlinear mapping relationship between the image and additional information with acoustic absorption spectra. At this stage, desired sound absorption spectra are input into the trained CGAN, and appropriate MMPP patterns will be predicted by the G network.

4.1. Image post-processing

By inputting a target sound absorption curve in a 1×75 dimension format into the trained model, an 88×88 MMPP structural configuration conforming to this absorption characteristic will be generated. Most of the directly generated structural configurations are difficult to fabricate in solid form and therefore require post-processing of the original image. This is because the 88×88 pixels pattern has a high degree of freedom, and most of the generated patterns contain small isolated pixels with extremely thin features, as enclosed by the blue dotted line in Fig. 10, which is not conducive to practical production. In addition, due to the arbitrary nature of image generation, jagged edges at material boundaries, such as areas surrounded by the pink dashed line in Fig. 10 will appear, increasing manufacturing difficulty and cost. Therefore, there are two main steps in image post-processing: one is filtering out redundant pixel points, and the other is smoothing the material boundaries.

To evaluate the sensitivity of acoustic performance to geometric perturbations introduced during post-processing, the sound absorption characteristics of three randomly selected original and post-processed patterns are compared. Fig. 10(a) illustrates a generated pattern with partitions, Fig. 10(b) presents a generated pattern containing double C-shaped cavities, and Fig. 10(c) depicts a generated pattern with a rectangular-shaped cavity and partitions. The sound absorption spectra of the post-processing image and the

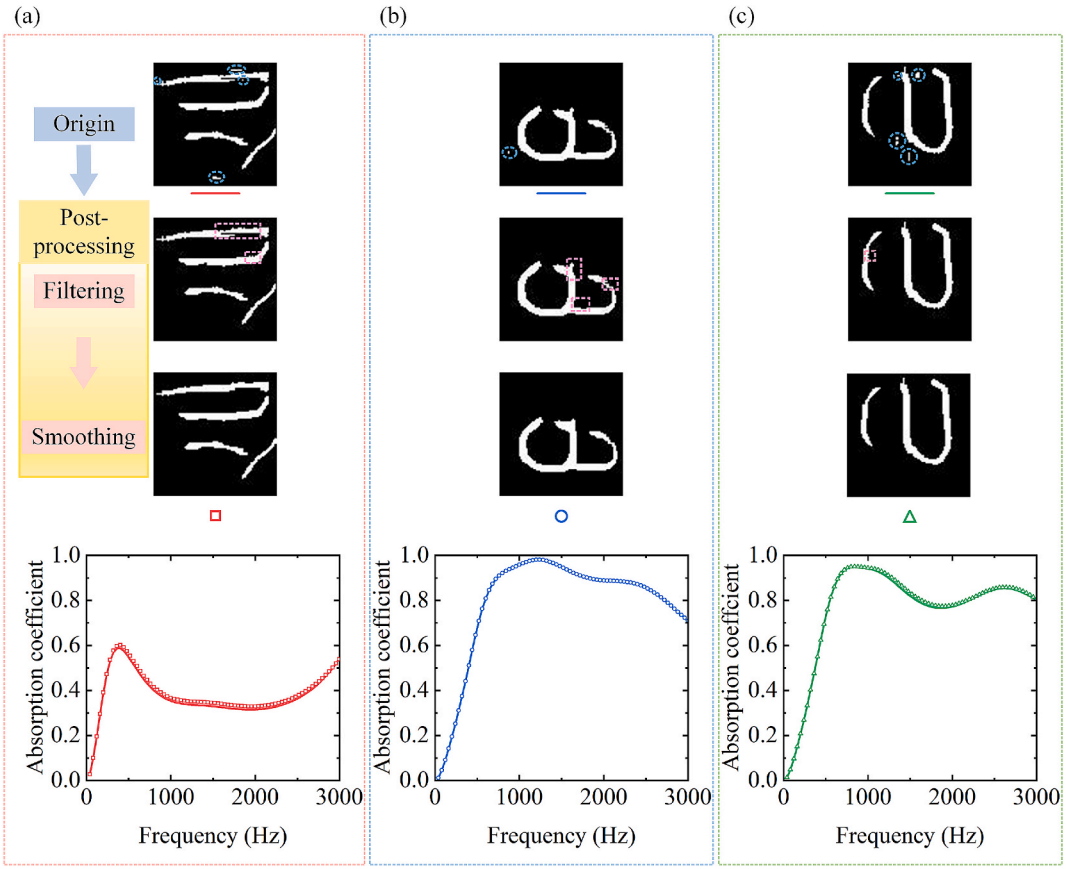


Fig. 10. Illustration of the post-processing procedure for the predicted patterns generated by the trained CGAN, along with a comparison of the sound absorption performance between the original and post-processed patterns: (a) partitions, (b) C-shaped cavity, and (c) rectangular-shaped cavity.

original ones are virtually identical, indicating that post-processing has a negligible impact on the overall acoustic performance.

4.2. Evaluation of trained CGAN on the test dataset

To quantitatively assess the inverse design performance of the trained model, the test dataset that is never used during the training process is input into the trained CGAN. There are 2,163 sound absorption spectra in the test dataset, which are set as desired absorption characteristics for generating the corresponding MMPP configurations. Subsequently, the 2,163 post-processed MMPP configurations are subjected to numerical simulation to obtain the predicted sound absorption spectra, which are then compared with the target absorption spectra.

Two error metrics are introduced to measure the accuracy of inverse design by a trained model: mean squared error (MSE) and mean absolute percentage error (MAPE), which are described by

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\alpha_p - \alpha_d)^2, \tag{20}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{\alpha_p - \alpha_d}{\alpha_d}\right|, \tag{21}$$

where α_p is the predicted absorption spectra calculated by simulation, and α_d are desired absorption spectra. This combined evaluation helps eliminate reliance on a single metric, avoiding an imbalance in error assessment caused by focusing solely on absolute or relative values during model tests.

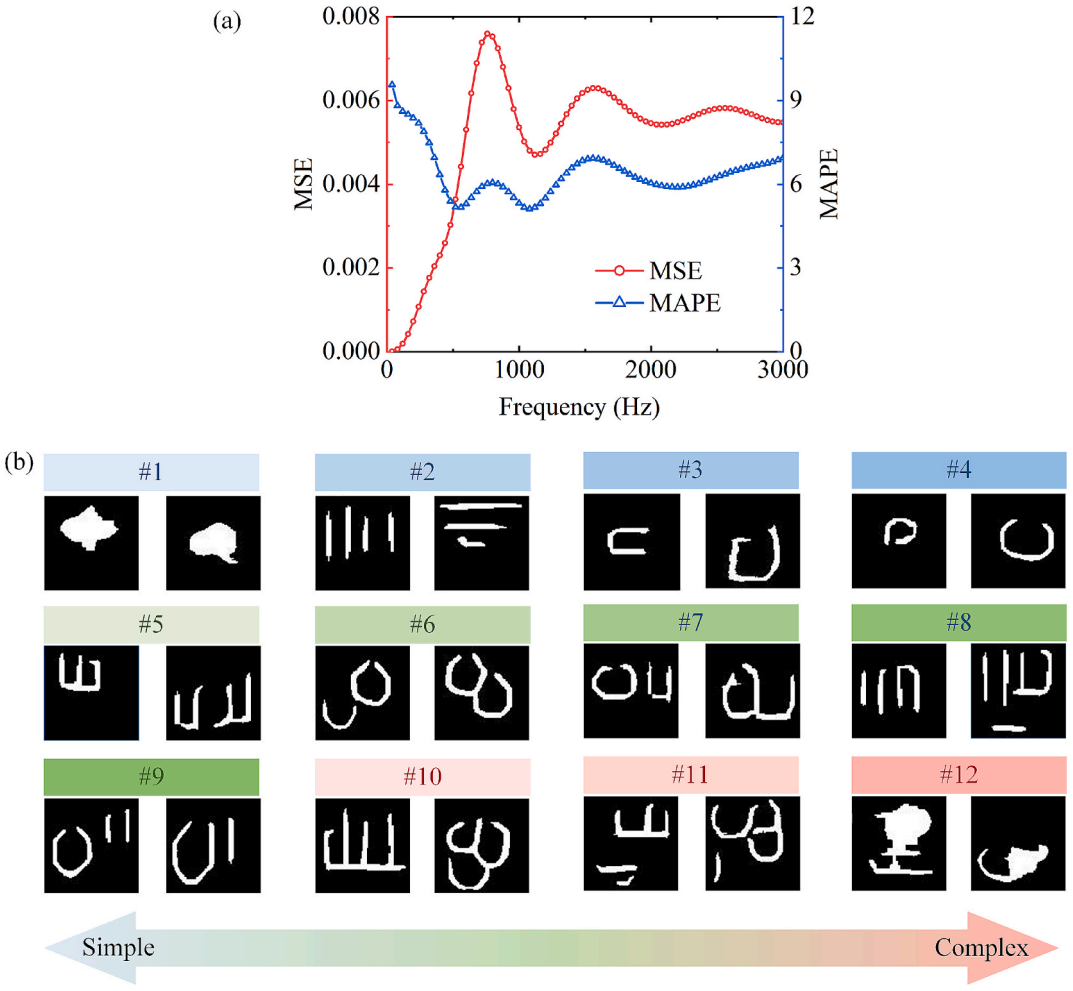


Fig. 11. (a) MSE and MAPE between the desired absorption spectra from the test set and predicted absorption spectra of MMPP generated by the trained model; (b) Typical types of patterns generated by the trained model.

If the MSE and MAPE approach zero, this means that the predicted values are in perfect agreement with the desired values. The MSE and MAPE between 2,163 expected values and corresponding predicted values are shown in Fig. 11(a). As depicted in the MSE and MAPE curves, the maximum MSE occurs at 740 Hz with a value of 7.59×10^{-3}, while the maximum MAPE appears at 40 Hz, reaching 9.56%. Over the considered frequency range, the MSE and MAPE remain at approximately 5.03×10^{-3} and 6.39%, respectively. Although the MAPE is relatively high at low frequencies, the corresponding low value of MSE at the same range suggests that this is primarily due to the low absorption coefficients of the target spectra in the low-frequency range (below 500 Hz).

Overall, the deviation between the predicted sound absorption spectra and the desired sound absorption spectra for the generated MMPP configuration is within an acceptable range, indicating that the trained model achieves high accuracy.

Furthermore, the pattern effectiveness and diversity of MMPP configurations generated through trained CGAN need to be emphasised when evaluating the inverse design performance of the trained CGAN. Notably, the original dataset does not contain patterns representing homogeneous porous materials without inclusions. Therefore, any inversely designed pattern resembling a homogeneous porous material is considered an invalid design. To quantify the effectiveness of the inverse design, the pattern-effectiveness metric is introduced and defined as (total number of designs − number of invalid designs) / (total number of designs). Certain generated patterns contain scattered small pixel blocks, which form homogeneous porous materials after post-processing and are thus regarded as invalid designs. There are only two patterns after post-processing without any rigid inclusions.

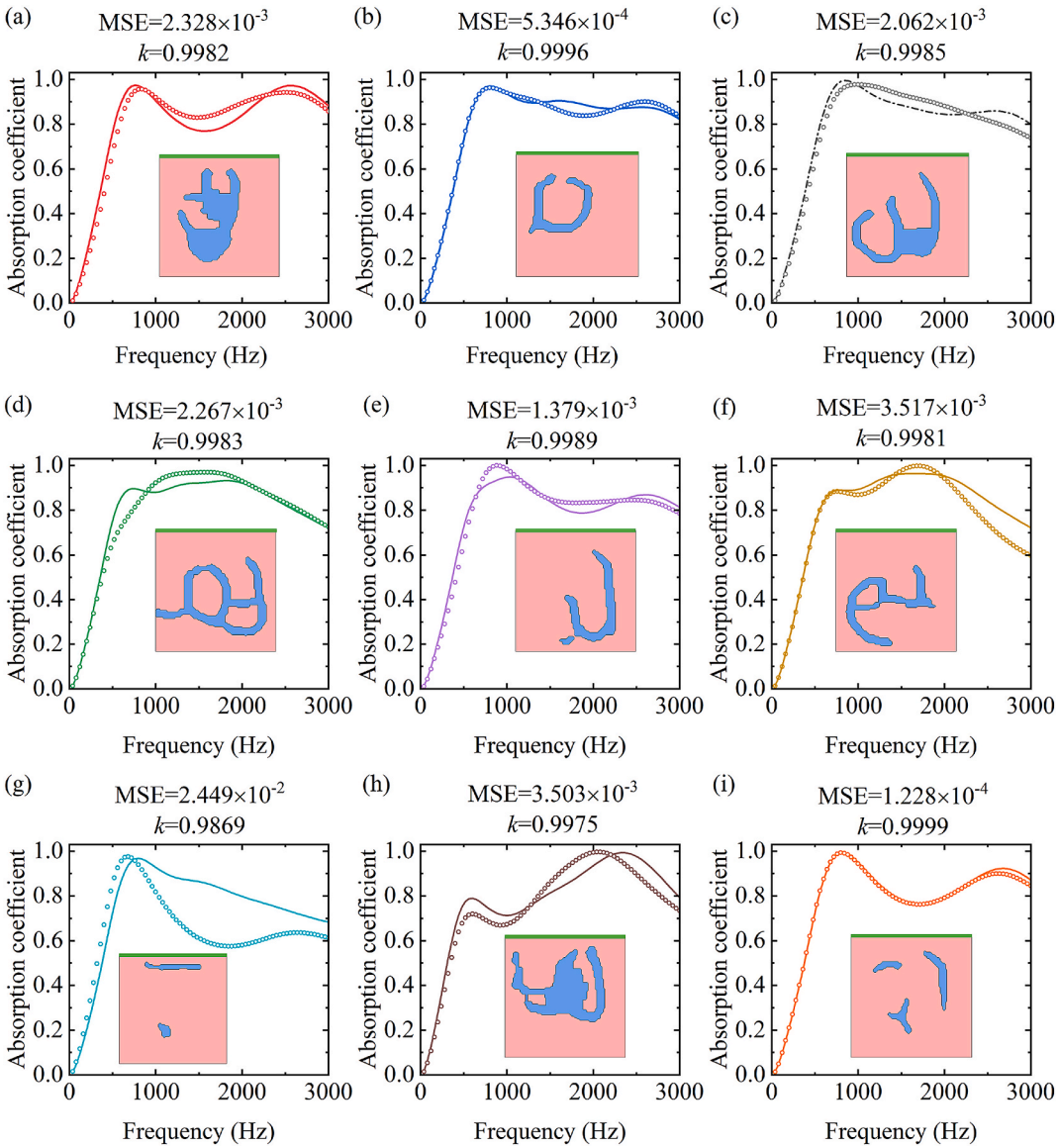


Fig. 12. Comparison between prediction sound absorption spectra of patterns designed by the trained model and sound absorption spectra selected from the test set.

Consequently, the pattern-effectiveness is calculated as (2,163–2) / 2,163 = 99.91%, demonstrating that the inverse design of MMPP guided by CGAN has high pattern-effectiveness.

As depicted in Fig. 11(b), among the 2,163 designs obtained, 12 typical pattern types for MMPP are proposed by the trained model. These pattern types are mainly composed of four basic elements: closed elements, partitions, rectangular-shaped cavities, and C-shaped cavities.

Overall, all designs can be classified into two categories:

The first category consists of patterns directly derived from the original dataset. Specifically, patterns #1-#4 show a single inclusion embedded in porous materials, and patterns #5-#9 show double inclusions embedded in porous materials;

The second category comprises novel configurations that creatively combine basic elements in the original dataset (#10-#12). These creative types can be described as follows:

Patterns #10: Multiple (more than two) thin-walled cavities with different opening directions;

Patterns #11: Multiple (two or more) thin-walled cavities combined with partitions;

Patterns #12: Thin-walled cavities combined with closed-rigid inclusion element.

A clear positive correlation exists between the complexity of the patterns and the number of inclusions. The generated patterns not only inherit the characteristics of patterns from the dataset but also undergo creative evolution on this feature, which means that CGAN effectively extracts, processes, and recombines image features to generate diverse and innovative configurations.

Notably, in the original dataset, all thin-walled cavities have their openings oriented towards the incident sound waves, corresponding to the perforated plate position. However, the patterns generated by trained CGAN exhibit a broader range of inclusion orientations, as illustrated in Fig. 11(b), such as type #3, #4 and #8. This variation in opening direction plays a crucial role in modulating acoustic behaviour, further validating the adaptability and flexibility of the CGAN-guided inverse design approach.

4.3. Inverse design performance for MMPP

In order to better observe the image configurations corresponding to each absorption curve predicted by the trained CGAN, as well as the discrepancy between the predicted absorption curves and the input absorption curves, nine sound absorption spectra are randomly selected from the test set, which is never used during training. The inverse design performance of the trained model is then verified.

To comprehensively evaluate the performance of the target and predicted curves, this study employs multi-dimensional quantitative metrics. First, MSE is utilised to assess the point-wise numerical dispersion between the two curves, quantitatively measuring the numerical accuracy of the predictions. Secondly, cosine similarity k is introduced as an auxiliary metric to quantitatively analyse the overall shape similarity of the curves, which is described as

$$k(\alpha_p, \alpha_d) = \frac{\langle \alpha_p, \alpha_d \rangle}{\|\alpha_p\| \cdot \|\alpha_d\|} \tag{22}$$

where $\langle \alpha_p, \alpha_d \rangle$ represents the inner product of two absorption curve vectors, and the $\|\alpha_p\| \cdot \|\alpha_d\|$ denotes regular product of the two vectors. The value of k close to 1 denotes high similarity. The combination of these two indicators provides a more comprehensive and in-depth approach to curve comparisons.

Nine predicted absorption curves of MMPP generated by the trained model are compared with the prescribed absorption curves presented in Fig. 12. It is observed that the similarity values of samples (a)-(i) are all close to 1, and the MSE values are approximately 0. In particular, the predicted sample (i) is in complete agreement with the desired absorption characteristics, with a cosine similarity of 0.9999 and MSE value of 1.288×10^{-4}.

Among the nine samples, except for sample (g), which has a relatively slightly larger MSE, all the prediction curves show a high degree of agreement with the target curve. However, the two curves of the sample (g) have obvious morphological similarities, both of which show an absorption peak around 500 Hz, followed by a similar decreasing trend. The quantitative curve similarity k analysis further verifies their consistency. Minor discrepancies may result from the inherent complexity of the nonlinear data distribution and the stochastic nature of the generation process, which depends on random noise inputs; small variations in the noise can slightly influence the generated outputs. An analysis of the overall test set in Section 4.1 and the remaining eight samples demonstrate that the trained CGAN closely approximates the target distribution in the majority of generated samples. The trained model has demonstrated robustness and reliability, as evidenced by its ability to maintain a high degree of fidelity to the target distribution despite individual sample deviations. In addition, CGAN indirectly affects error performance through global distribution optimisation, and individual errors reflect more the intrinsic characteristics of the model design rather than the inherent limitations of the model itself.

4.4. Implicit learning of physical features by CGAN

In order to enhance the physical controllability and interpretability of the inverse design process, the implicit physical characteristics of the candidate designs generated by CGAN are further explored after the evaluation of the accuracy and design diversity of the CGAN model. A double-peak sound absorption curve with sound absorption peaks near 700 Hz and 2400 Hz is used as the target, the corresponding structures are inversely designed by CGAN. This type of double-peak curve is further divided into three types: (1) The first type: the first absorption peak (700 Hz) with $\alpha > 0.8$, the second absorption peak (2400 Hz) with $\alpha < 0.8$, representing the low-frequency sound absorption design; (2) The second type: both absorption peaks with $\alpha > 0.8$, representing the broadband sound

absorption design; (3) The third type: the first absorption peak with $\alpha < 0.8$, the second absorption peak with $\alpha > 0.8$, representing the high-frequency sound absorption design.

The above three types of target curves are input into the trained CGAN network, respectively, and three candidate designs are generated for each type of curve. Taking the first type of curve as an example, the corresponding three groups of structures and predicted sound absorption performance are illustrated in Fig. 13(a). It can be observed that the sound absorption curve predicted by the candidate structure is highly consistent with the target curve. The relative MSE between the three groups of samples and the target is small, and the cosine similarity is close to 1. The second and third types of designs shown in Fig. 13(b) and (c) also show good matching with the target, further confirming the design reliability of CGAN.

Furthermore, the three types of designs are analysed from the perspective of structural configuration and material distribution, revealing the physical differences between the designs.

For the first type of low-frequency sound absorption design, as shown in Fig. 13(a), the inclusions are short and small partitions, mainly distributed in the shallow surface area of the porous materials, and the central area is marked with a grey dashed line. The inclusion filling rates ϕ_R are 1.4%, 1.1% and 3.8%, respectively. The overall filling rate is low, showing the characteristics of low density and shallow embedding.

For the second type of full-frequency optimisation design, as illustrated in Fig. 13(b), the rigid inclusion forms are diverse,

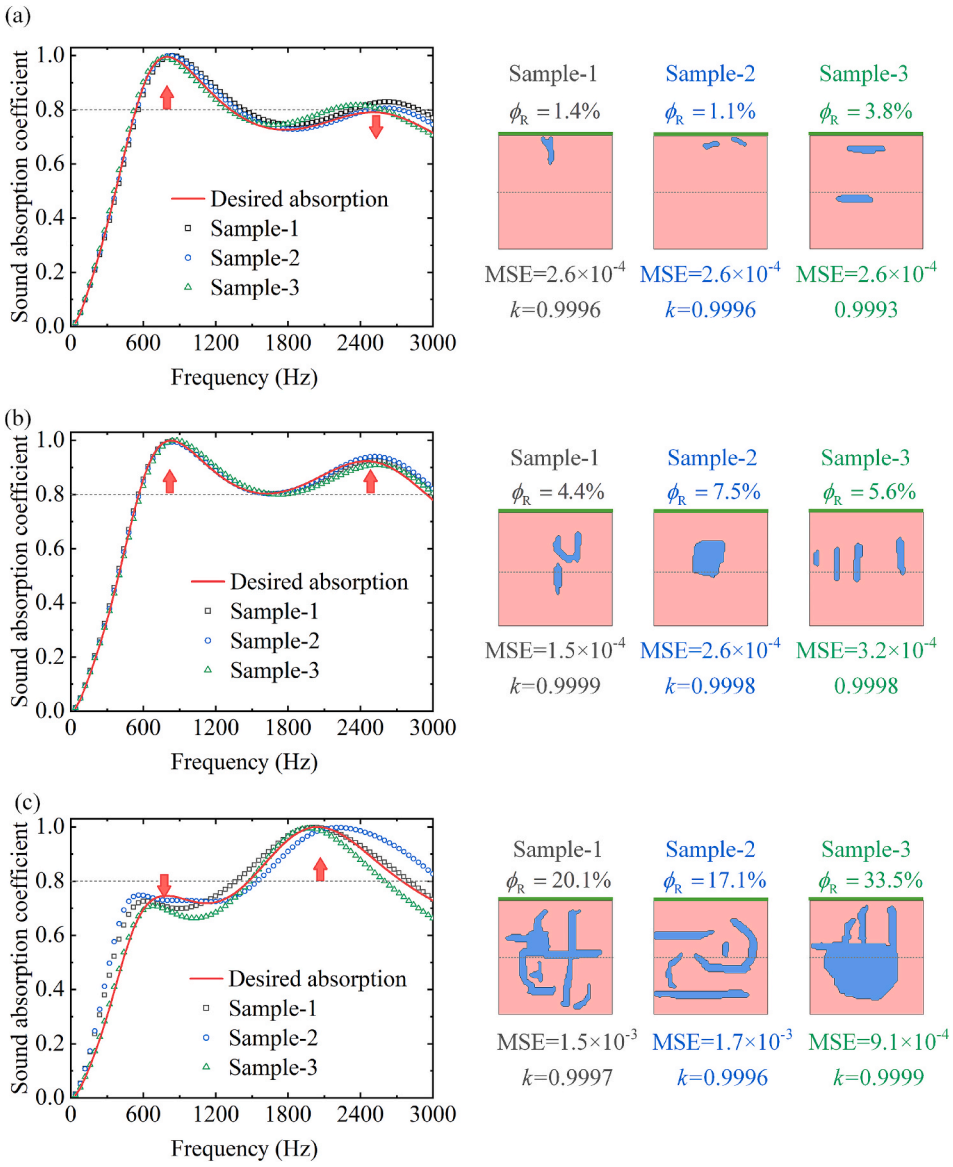


Fig. 13. Structural configurations generated by CGAN and the corresponding predicted sound absorption curves for (a) desired absorption-1, (b) desired absorption-2, (c) and desired absorption-3.

including thin-walled C-shaped cavities, closed rectangular inclusions, and multi-partition structures. Compared with the first type, its inclusion is closer to the central area, and the filling rate is significantly increased, reflecting the layout characteristics of medium density and central concentration.

Moreover, for the third type of high-frequency sound absorption optimisation design, as displayed in Fig. 13(c), the inclusion is large in volume and deeply distributed, with filling rates of 20.1%, 17.1% and 33.5%, respectively. The inclusion shapes are innovative combinations of partitions, C-shaped cavities, rectangular cavities and closed inclusions. These configurations are not present in the original dataset and exhibit characteristics of high structural density and deep embedding.

The physical principles underlying the candidate designs can be summarised as follows: when targeting low-frequency sound absorption, the model tends to generate structures with small-sized inclusions, shallowly embedded and sparsely distributed. In contrast, designs aimed at high-frequency absorption are characterised by large-volume, complex inclusion combinations with a high filling ratio, often embedded deeper within the porous materials. For broadband absorption, the design is characterised by a moderate filling rate, diverse inclusion morphologies, and a spatial distribution concentrated around the middle region.

This result shows that the CGAN model can not only achieve accurate inverse design, but also has the ability to explore the implicit relationship between sound absorption characteristics and structural configurations, providing designers with a knowledge-guided structural design basis under data-driven, allowing designers to intuitively grasp the inherent physical laws between performance and structure.

4.5. *Customised low-frequency broadband absorption of MMPP*

In summary, the inverse design method for customized MMPP using CGAN architecture has five distinct yet interrelated steps, as depicted in Fig. 14: (1) defining target absorption spectra according to the needs of the application scenario; (2) inputting target absorption spectra into trained CGAN; (3) post-processing the inverse-designed MMPP configuration to ensure practical feasibility; (4) validating the sound absorption performance of the customized MMPP configuration against the target absorption curve using the FE model and experimental verification; and (5) selecting the configuration with the smallest error relative to the target curve or the most fabrication-friendly design from multiple generated samples for implementation.

These steps are closely interconnected, each contributing to the overall success of the method. For example, to achieve low-frequency broadband absorption, the target absorption curve is defined to reach an absorption coefficient of 0.5 at 360 Hz and 0.8 above 560 Hz, as depicted in Fig. 15(a). Using this target, this method generates four candidate designs, shown in Fig. 15(b)–(e), with their corresponding absorption spectra compared to the target spectra in Fig. 15(a).

The results demonstrate that CGAN effectively produces diverse inverse design solutions capable of meeting the specified sound absorption targets. Notably, even when initialised with the same target curve, the trained CGAN generates a variety of design alternatives, offering flexibility and multiple options for practical application.

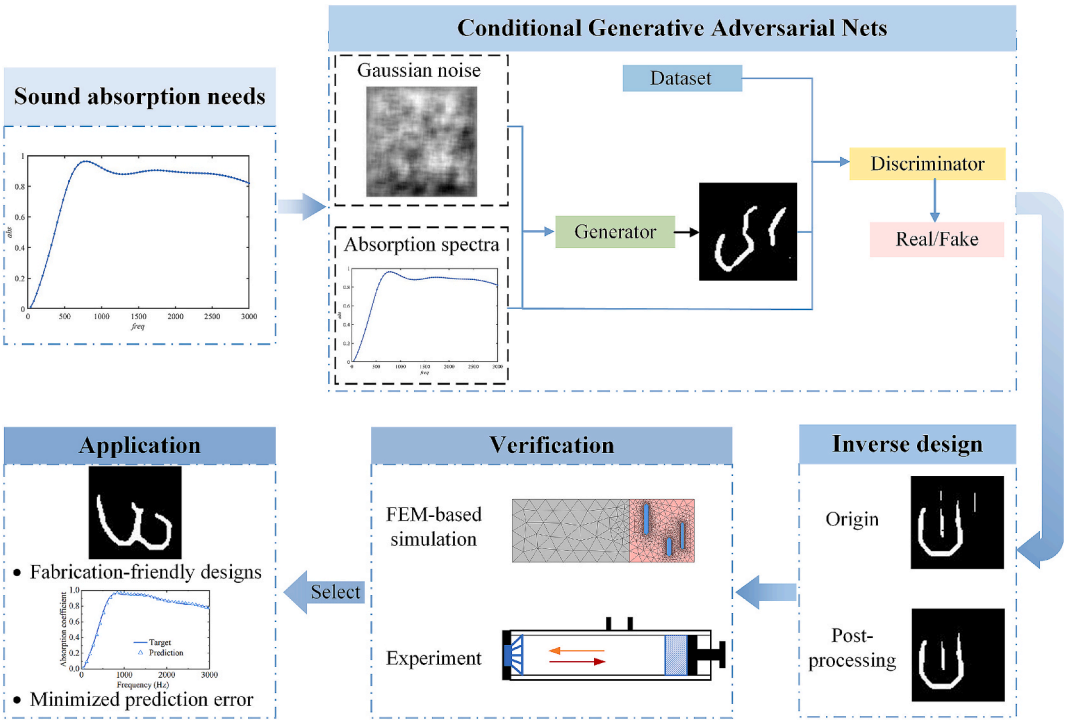


Fig. 14. Schematic of the on-demand inverse design process for MMPP based on CGAN.

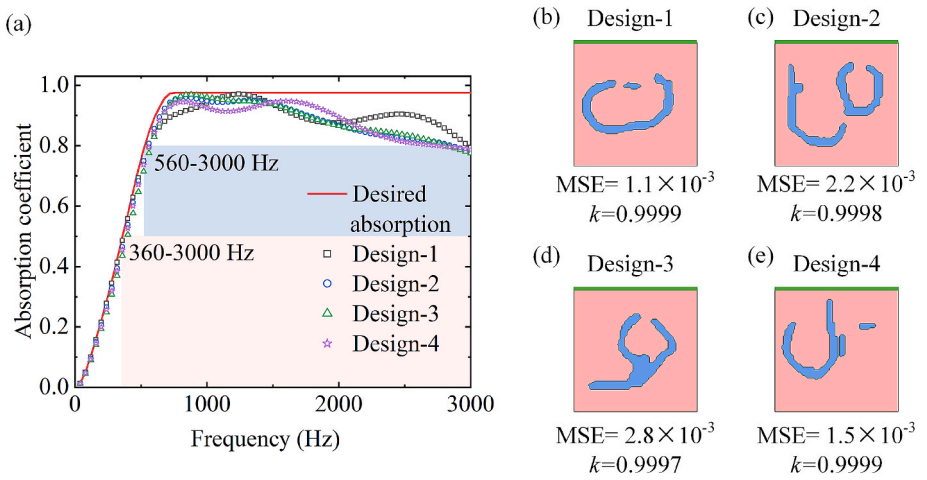


Fig. 15. (a) Target absorption spectra and predicted absorption spectra of four designs proposed by the trained model with the additional message of target absorption spectra; Schematic of four inverse designs for MMPP: (b) Design-1, (c) Design-2, (d) Design-3, and (e) Design-4.

Compared to topology optimisation, which relies on physical laws and iterative solving of complex equations to define and optimise objective functions, the CGAN-based inverse design method offers a data-driven alternative. Topology optimisation requires a lot of calculations for each design iteration, while CGAN involves only a one-time training phase. Once trained, the model can generate designs in seconds. As a complementary approach, it strikes an effective balance between computational efficiency and acceptable accuracy.

Although the initial dataset collection and model training require significant computational time, the CGAN-based inverse design method is able to generate candidate solutions for the target performance curve almost instantly. Users only need to input the target curve to quickly obtain candidate solutions that meet the target performance criteria. This method can be reused for similar noise control design problems with target in the 40–3000 Hz range, significantly simplifying the metamaterial design process and reducing the time and cost typically associated with iterative trial-and-error approaches.

Rather than replacing expert knowledge, this data-driven approach serves as a powerful assistive tool that allows researchers to explore complex design spaces more efficiently and systematically. It also facilitates the rapid discovery of non-intuitive yet physically meaningful solutions, accelerates early-stage design exploration, and supports the accumulation of design knowledge and physical intuition, particularly for less experienced users.

Importantly, all candidate designs generated by the model must undergo subsequent physical evaluation and validation to ensure feasibility and consistency with physical principles, ensuring that domain expertise guides the final design process.

4.6. Sound absorption mechanism of MMPP

To figure out the sound absorption mechanism of four MMPP configurations designed inversely by the trained model, the typical Design-1 with a minimum MSE is chosen as a representative case for further analysis. Firstly, explore the difference in acoustic performance between the 2D FE model and the three-dimensional (3D) FE model. A 3D FE model with an entity perforated plate with the same geometric parameters as the Interior Perforated Plate in the 2D model is established. The sound absorption spectra of Design-1, calculated by the 2D FE model and 3D FE model, are compared in Fig. 16(a). It can be seen that the calculation results of the two model settings are basically the same.

The 2D FE model cannot intuitively simulate the effect of perforated plates on the sound field. Hence, a 3D FE model is used to gain deeper insights into the underlying mechanism of MMPP designs. Fig. 16(b)–(e) show the absolute relative sound pressure ($|p/p_0|$) distribution and sound velocity (v) arrows of Design-1, Design-2, Design-3 and Design-4 at the non-resonant frequency ($f0$) and three absorption peaks (P1, P2 and P3), respectively, where p_0 is the incident sound pressure.

These four design structures have similar sound absorption characteristics, and the sound pressure field and sound velocity field characteristics they present have common characteristics. It is observed from Fig. 16(b)–(e) that the variation in the sound pressure field distribution inside the MMPP at absorption peak frequencies (P1, P2 and P3) is significantly greater than that at the non-resonant frequency ($f0$). Obviously, the sound velocity exhibits a maximum value at the entrance of the perforated plate, which indicates that the perforated plate facilitates enhanced energy dissipation.

Moreover, at P1, the sound pressure is mainly distributed in the entire porous material area behind the perforated plate, indicating that the sound absorption mainly comes from the composite cavity structure composed of the perforated plate and the porous material behind it. At this time, the sound wave passes through the perforated plate and can be effectively coupled to the absorption layer behind it. The air mass at the perforated plate orifice undergoes mass-spring resonance, forming energy accumulation; the excited sound energy is then effectively attenuated through the viscosity and heat dissipation mechanism of the porous material, thereby achieving the synergistic low-frequency absorption of the perforated plate and the porous material.

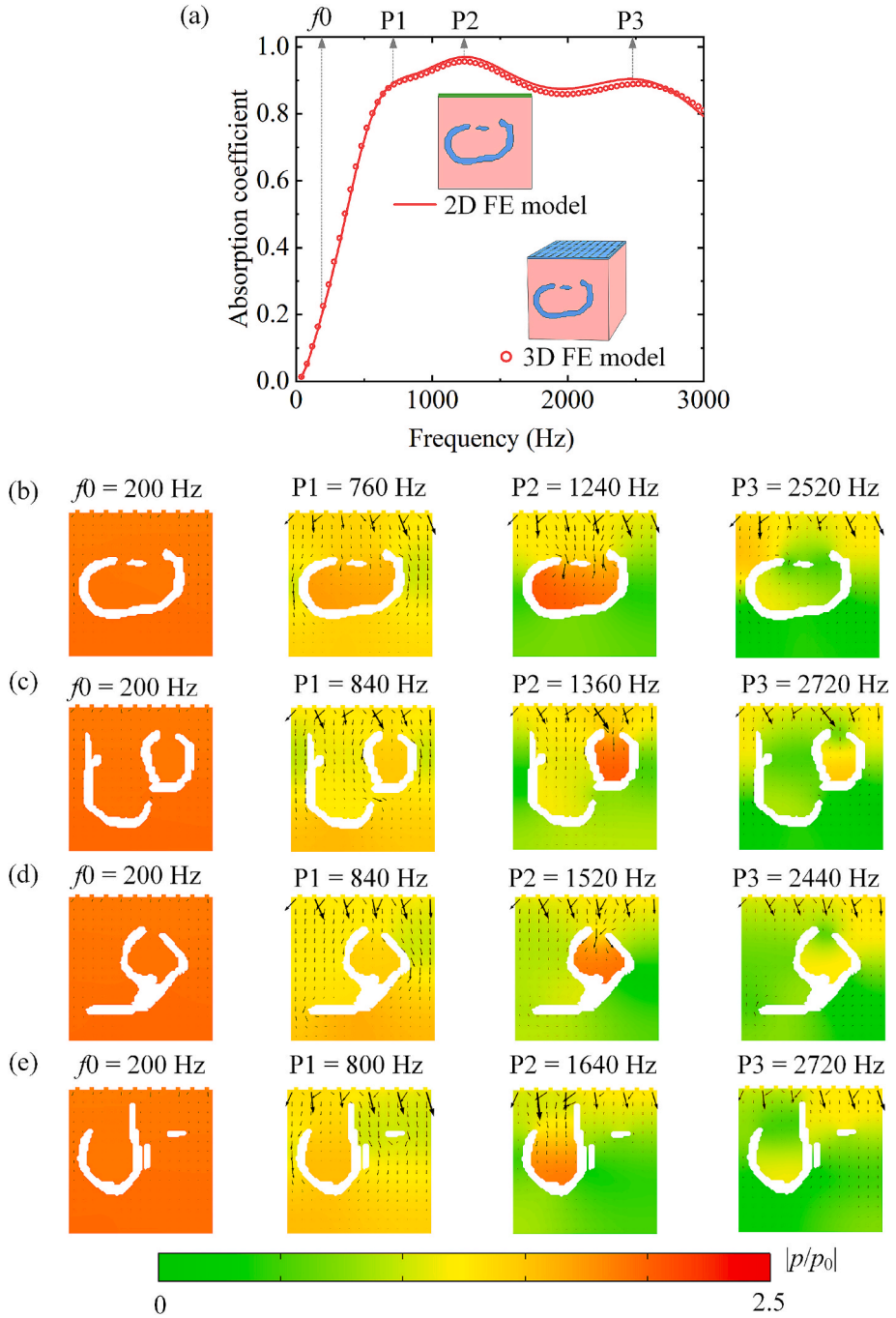


Fig. 16. (a) Comparison of sound absorption coefficient between 2D FE model and 3D FE model for Design-1; Sound pressure field distribution and sound velocity arrows inside (b) Design-1, (c) Design-2, (d) Design-3, and (e) Design-4 at the non-resonant frequency *f*0, the first sound absorption peak P1, the second sound absorption peak P2, and the third sound absorption peak P3.

The mechanism corresponding to the P2 absorption peak is a composite local resonance system formed by thin-walled rigid inclusions in the MMPP (such as C-shaped cavities, rectangular cavities, partitions, etc.). When the sound wave enters the structure, the sound pressure and sound velocity vectors are focused into the inclusions. This indicates that the sound wave at this frequency excites the local resonance mode induced by the inclusions, and the local energy concentration causes the amplitude of the sound velocity of the porous material near the inclusions to increase, thereby improving the dissipation capacity. This local resonance mode of inclusions causes the sound wave to gradually dissipate in the MMPP, thereby achieving strong absorption.

The sound pressure at P3 is primarily concentrated at the top of the porous region and along the boundaries of the inclusions,

forming a complex localised sound field distribution. The acoustic velocity vectors in these regions exhibit dense perturbations and backpropagation, indicating pronounced local resonances. Due to the shorter wavelength of high-frequency sound waves, multiple scattering and interference readily occur at the inclusion boundaries. Under the coupling of higher-order modes, the viscous and thermal dissipation mechanisms within the porous materials are effectively activated. Moreover, the more complex the geometry of the inclusions, the more tortuous the propagation paths become, further enhancing energy confinement and dissipation effects.

In summary, design structures generated by CGAN not only achieve accurate fitting of the target sound absorption curve, but also demonstrate that the structure–performance mapping learned by the model exhibits strong physical consistency, as discussed in Section 4.4. Specifically, low-frequency targets tend to generate shallow and small-sized inclusions, as their sound absorption is primarily governed by the resonance formed between the perforated plate and the porous cavity. In contrast, high-frequency targets correspond to deeply embedded and larger inclusions, where sound absorption is mainly dominated by higher-order resonance modes and multi-path interference occurring near the tops and boundaries of the inclusions. The physical controllability revealed by the deep learning model thus provides a theoretically grounded and physically interpretable basis for the design of acoustic metamaterials.

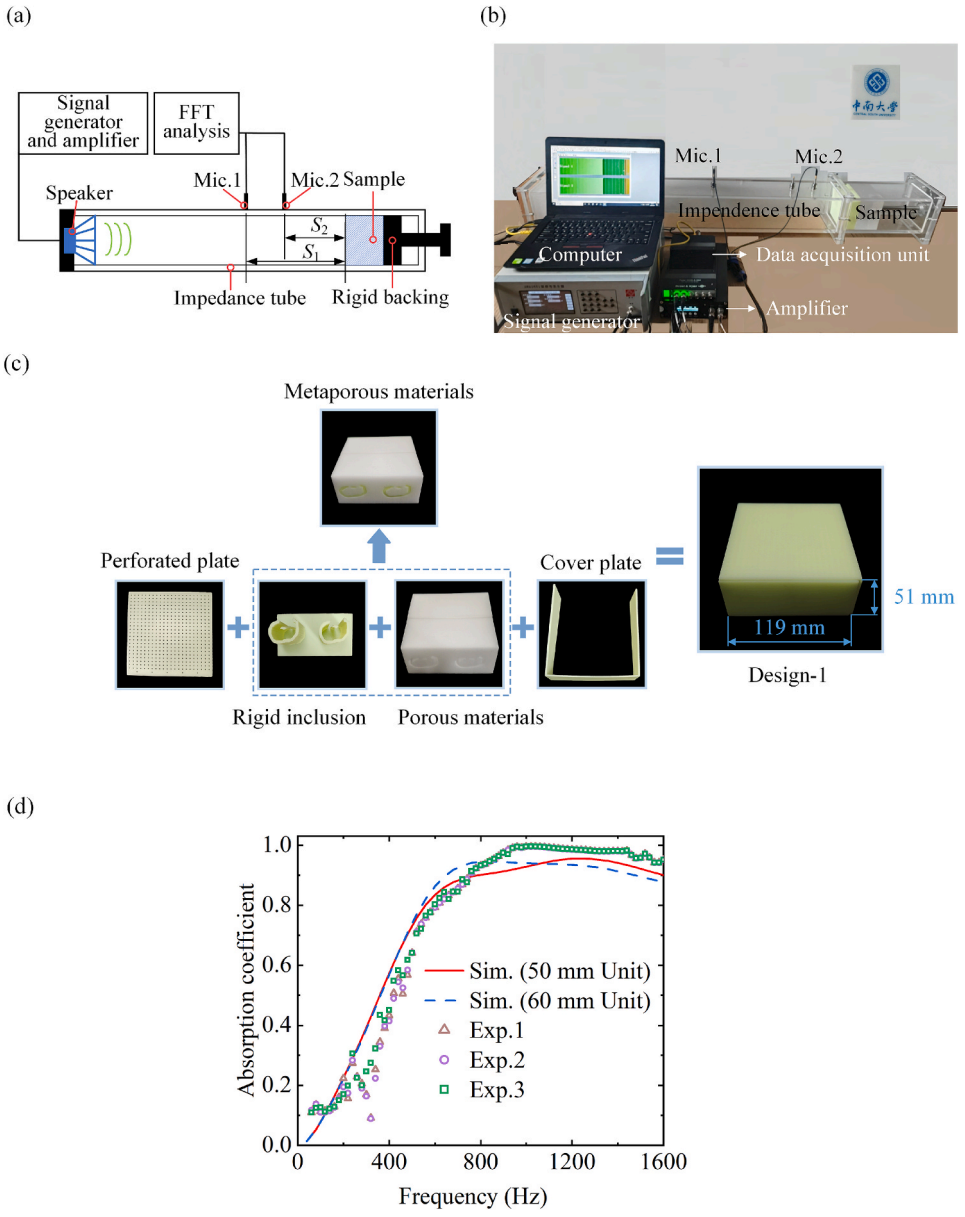


Fig. 17. (a) Experiment principle of absorption coefficient measurement using the two-microphone method; (b) Experiment environment and experiment setup; (c) Assembly process of experimental sample; (d) Comparison of simulation results of Design-1 with 50 mm unit and 60 mm unit and experiment results of Design-1.

5. Experiment verification

To verify the manufacturability and effectiveness of inverse-designed MMPP, Design-1 was selected as the experimental configuration and manufactured. The corresponding sound absorption coefficient of Design-1 was measured by a square impedance tube with a measurement frequency range of 60–1600 Hz, based on the two-microphone method. The experimental principle is displayed in Fig. 17(a), and the corresponding experimental equipment and environment are presented in Fig. 17(b).

The square impedance tube with a side length of 120 mm, so at least two units of Design-1 are required for testing. Given that the dimensions of the double units of Design-1 are 100×100 mm^2, each unit was extended by 10 mm along the x-direction. To facilitate easy placement and removal within the impedance tube, the actual sample size was set to 119×119 mm^2, slightly smaller than the tube's cross-sectional size. During the production of the experimental sample, a hard plate with a wall thickness of 2 mm was introduced around the porous material matrix to support and secure the perforated plate. As a result, the effective absorption material size of the experimental sample was 115×115 mm^2. The assembly process of the experimental sample is exhibited in Fig. 17(c).

The physical fabrication of proposed acoustic metamaterials mainly consists of four components: a perforated plate at the top, a rigid cover plate around the edges, a porous material matrix, and rigid inclusions. Among them, the perforated plate, cover plates and rigid inclusions were made of resin, which was 3D-printed using SLA technology with a printing accuracy of 0.01 mm. The porous materials were melamine foam purchased from *Xuxian Soundproofing and Insulation Building Materials* Store, and they were formed into a specific shape by stamping. These components were assembled to construct the designed acoustic metamaterial for experimental evaluation.

The sound absorption spectra of the extended unit are slightly different from those of the original unit of Design-1. However, since the dimensions of the embedded inclusions and the thickness of the porous material remain unchanged, the overall sound absorption trend is similar to that of the original Design-1, as shown in Fig. 17(d). To avoid errors caused by improper operation, repeated experiments were conducted three times, and the experimental results are shown in Fig. 17(d).

It is reflected that the experiment results are consistent with the simulation results, which validates the accuracy of the numerical analysis. The deviation between experiment results and simulated results may come from manufacturing tolerances in the 3D-printed parts, as well as imperfect fits during assembly. Additionally, the introduction of the cover plate used to support the experimental sample may induce a frequency shift in the measured absorption spectra.

6. Conclusions

In summary, an inverse design method based on CGAN for metaporous materials covered with a perforated plate is proposed, aiming to broaden the structural design space and achieve efficient broadband sound absorption. CGAN is mainly used to guide the design of inclusions in metaporous materials. The MMPP in the dataset is generated by a control function and random parameter combinations under physical constraints, guided by prior knowledge to ensure the diversity in design patterns and extensiveness in sound absorption data.

The CGAN-based inverse design method demonstrates modularity and reusability, enabling fast, spectrum-driven exploration in a high-degree-of-freedom space without repetitive simulations. The accuracy and effectiveness of the trained CGAN model are validated with a test set, yielding an MSE of 5.39×10^{-3} and an image design efficiency of 99.1%. This design approach can also create innovative structures based on and beyond the configuration of the data set. Further observation of the relationship between the absorption curve characteristics and the generated structural features revealed that low-frequency absorption is related to small-sized, shallowly embedded inclusions, while large-sized, deeply embedded inclusions are conducive to high-frequency absorption. Unlike existing studies focusing solely on peak absorption enhancement, this method offers the ability for user-defined spectral shaping. For target sound absorption performance within the effective frequency range of 360–3000 Hz, multiple predicted designs are generated and validated through both simulations and experiments, demonstrating a synergistic absorption mechanism involving micro-perforated panel resonance, local resonance of inclusions, higher-order resonant modes, and acoustic wave interference.

This inverse design method overcomes the computational limitations of forward design with high-dimensional parameters, enabling diverse and creative designs of specified physical properties, while considering manufacturability. More importantly, this method enables autonomous learning of structure-performance relationships, reflecting inherent physical laws such as optimising absorption in a given frequency band by adjusting inclusion size and distribution, which advances our understanding of the complex physics in acoustic metamaterials. This demonstrates that the inverse design method can serve as a design tool in academic research, assisting researchers in uncovering new insights into structure-performance relationships and enhancing understanding of material design, thus maintaining scientific control while improving design performance and efficiency.

CRediT authorship contribution statement

Jiahui Yan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yingli Li:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition. **Guohui Yin:** Writing – review & editing, Visualization, Software, Methodology. **Song Yao:** Writing – review & editing, Supervision. **Yong Peng:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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